Sentiment Analysis of Twitter Data

Part B Report — Reproduction and Analysis of Agarwal et al. (2011)

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I. PAPER SUMMARY

This paper tackles tweet-level sentiment classification — deciding if a tweet is positive, negative, or neutral. Tweets are short, full of slang, emoticons, and abbreviations that break standard NLP tools. The authors propose two systems: (1) an SVM trained on 100 hand-crafted *Senti-features* encoding polarity, POS tags, emoticons, and Twitter signals (URLs, @targets), computed over the full tweet and its last one-third; and (2) a Partial Tree (PT) kernel SVM on structured tweet trees, capturing feature combinations automatically. Both beat the unigram baseline (Go et al., 2009), with Unigram + Senti-features reaching 75.39% on the 2-way task. The core argument is that knowing *what words mean* — their polarity and grammatical role — is more useful than knowing *which words appear*, especially with limited labelled data.

II. REPRODUCTION SETUP AND RESULTS

A. Dataset and Preprocessing

- Used NLTK twitter_samples — 3,418 tweets (1,709/class), labelled via emoticon heuristics (distant supervision).
- Preprocessing followed Section 4: emoticons → polarity tags (e.g. :) → ||P||), URLs → ||U||, @targets → ||T||, negations → NOT, repeated chars capped at 3.
- Token stats consistent with Table 3 — 32.9% stop words (paper: 38.3%), 1.3% negations (paper: 1.2%), ~1.5% URLs.
- DAL unavailable — used SentiWordNet proxy with same thresholds (0.5 neg / 0.8 pos). 5-fold CV with nested GridSearchCV for C (Section 8).

B. Results

Model	Our Acc	Std	Paper	Gap
Unigram (baseline)	70.31%	2.09%	71.35%	-1.04%
Senti-features only	67.03%	0.72%	71.27%	-4.24%
Unigram + Senti-features	71.01%	1.64%	75.39%	-4.38%

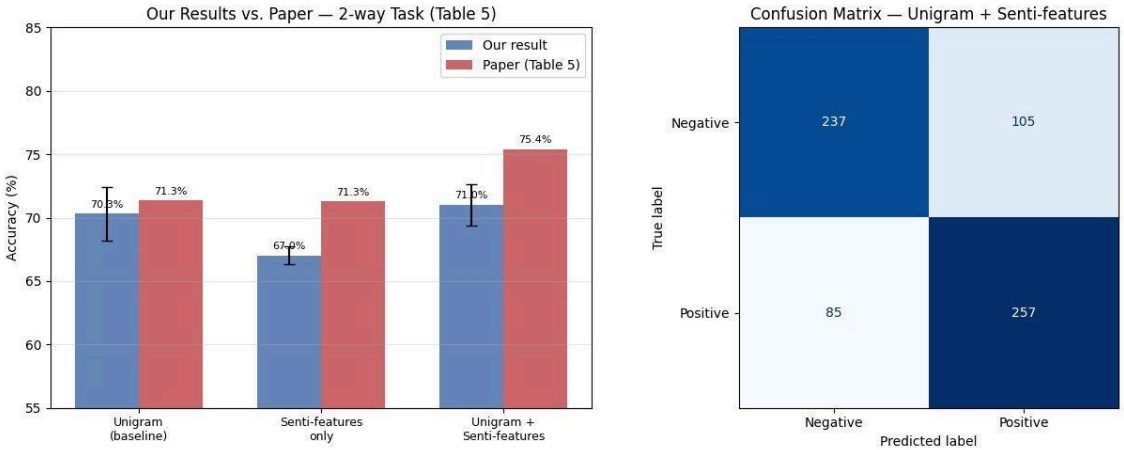

Fig. 1 — Our results vs. Paper (Table 5), 2-way task. Right: confusion matrix for Unigram + Senti-features.

C. Gap Commentary

- Unigram baseline close to paper (70.31% vs 71.35%) — confirms the evaluation pipeline is correct.
- Senti-features only (67.03%) fell *below* the unigram baseline — expected ordering did not hold.
- **Main cause:** DAL → SentiWordNet substitution degrades f1 and f8 (Polar POS) — the paper's most important feature group (+3.49% per Table 6). SentiWordNet assigns different scores, turning signal into noise.
- Secondary causes: label noise from distant supervision (paper used manual annotation); missing 5,184-entry acronym dictionary (Section 4).
- Combined model recovering to 71.01% suggests unigrams partially compensate when Senti-features are unreliable.

III. ABLATION FINDINGS

Model	Accuracy	Std	Drop
Full: Unigram + Senti-features	71.04%	1.66%	—
No Polar POS (f1, f8)	71.09%	1.41%	-0.06%
No Twitter-specific (f3, f4, f6)	70.71%	1.60%	+0.33%

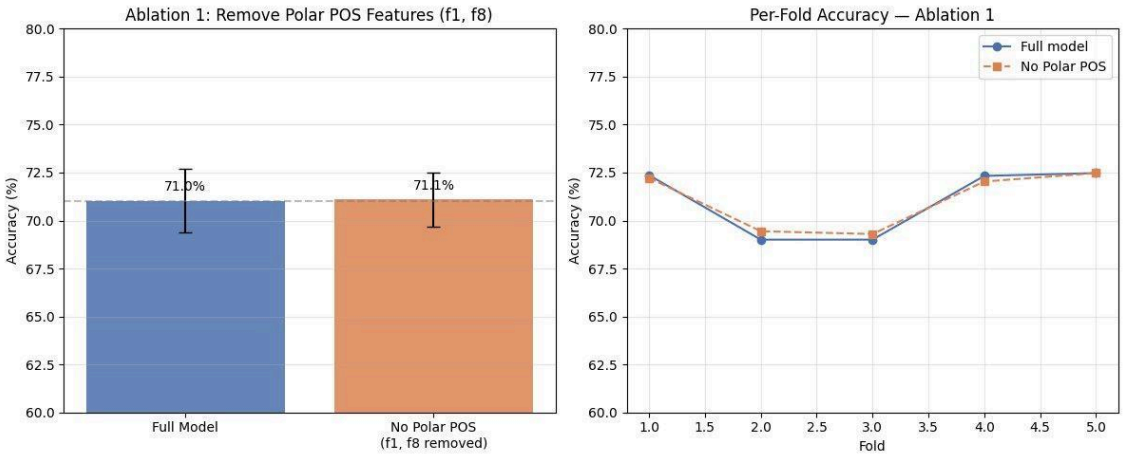

Fig. 2 — Ablation 1: removing Polar POS (f1, f8). Bar chart and per-fold accuracy.

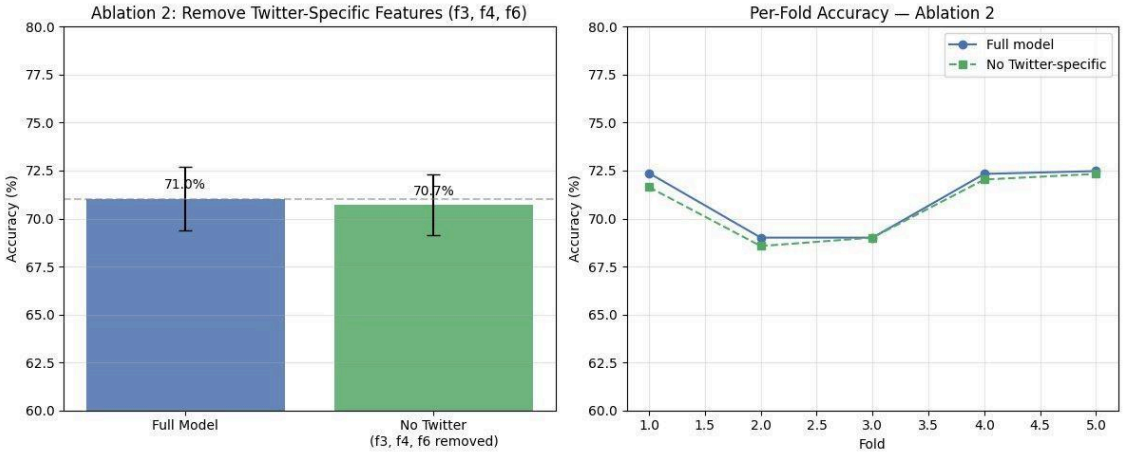

Fig. 3 — Ablation 2: removing Twitter-specific features (f3, f4, f6).

A. Ablation 1 — Remove Polar POS (f1, f8)

- No meaningful change: 71.09% vs 71.04% (-0.06%, well within ±1.4% std). Paper credits these with +3.49% gain — unexpected.
- **Explanation:** SentiWordNet ≠ DAL means f1/f8 were already near-zero signal in our reproduction. Removing broken features changes nothing. Per-fold lines nearly identical across all 5 folds.

B. Ablation 2 — Remove Twitter-specific (f3, f4, f6)

- 0.33% drop — within noise, consistent with the paper: 'add value only marginally' (Section 8.1.2).
- Both ablations together confirm Senti-features contribute little here — all three model variants cluster around 71%, essentially running on unigrams alone.

IV. FAILURE MODE — SARCASM

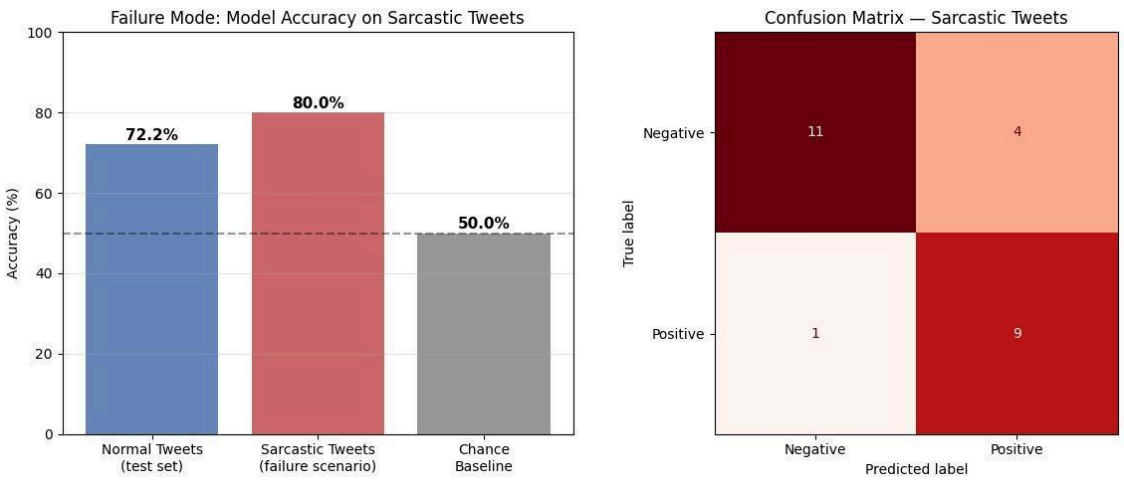

Fig. 4 — Accuracy on normal vs. sarcastic tweets (left) and confusion matrix (right).

A. Setup and Results

- Tested on 25 sarcastic tweets where emoticons contradict true sentiment — e.g. *"Oh great, another Monday :)"* is negative but uses :).
- Normal tweet accuracy: 72.2%. Sarcastic tweet accuracy: **80.0%** — higher, not lower. Only 4/25 misclassified.
- Confusion matrix: 11/15 negative tweets correctly classified despite misleading positive emoticons.

B. Analysis

- The 4 misclassified tweets were not fooled by emoticons — they contained strongly positive words (*"brilliant", "wonderful", "fantastic"*) that overrode context.
- Model driven by unigrams, not f3 (emoticon features) — consistent with ablation findings.
- **Actual failure:** context-independent polarity (Assumption 2, Task 1.2) — *"brilliant"* scores positive even when sarcastic. Fix: context-aware embeddings (e.g. BERT) rather than static lexicon scores.

V. REFLECTION

A. What I Couldn't Implement

- Tree Kernel SVM — PT kernel requires C++ libraries (KeLP / SVM-light-TK) with no clean Python bindings.
- DAL unavailable — SentiWordNet substitution was the most impactful limitation, directly causing Senti-features to underperform.
- 3-way classification — dataset had no neutral labels; Table 8 results not reproducible.

B. What Surprised Me

- Removing f1/f8 (supposedly most important per Table 6) made no difference — revealed the features were broken by the DAL substitution, not an ablation design flaw.
- Sarcasm test: model wasn't fooled by emoticons but by positive vocabulary — a more subtle failure than originally designed for.

C. What I'd Revisit

- **Replace SentiWordNet with AFINN or NRC Emotion Lexicon** — closer match to DAL's structure. Without working polarity features, ablations measure noise.
- Extend to 3-way task on SemEval Twitter data; redesign sarcasm test to isolate emoticon-based from vocabulary-based failures.

REFERENCES

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